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International AS and A-level

Chemistry 9620
CH05: Unit 5 Practical and synoptic
Report on the Examination

January 2025

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REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH05 – JANUARY 2025

The mean has continued a steady increase.

Some students answered well across the paper, but there were many who scored a large part of their marks in Section B.

QUESTION 01

01.1 Most students answered this well.

01.2 Two extrapolations were needed, even though the temperature before 4th minute was constant. Common errors were misreading the scale or extrapolating to the wrong time.

01.3 A subtle point, missed by some, is that the graphical method does not stop heat loss but rather allows for it. The non-zero response time of the thermometer was allowed, even though not a major problem in this experiment.

01.4 This question was answered well by over 90% of students.

01.5 Some students identified that one component was in excess but then failed to choose the amount of the one that would react completely. Some students calculated the difference.

QUESTION 02

02.1 Although almost all students recognised that impurities were being removed, only about 60% mentioned soluble impurities.

02.2 This was an example of students using a ‘good tips for practicals‘ answer rather than choosing the correct response. A water bath is, indeed, often used to maintain a steady temperature or control the speed of a reaction or to allow gentle heating but the important point here was that the ethanol was flammable.

02.3 About 60% mentioned insoluble impurities, despite almost all knowing that impurities were being removed.

02.4 Fewer than half the students scored this mark. Many wrote about faster cooling, constant temperature or to get better crystals.

02.5 Just over half the students scored both marks. Common errors included to get purer crystals without reference to removing more solvent and that the method is more efficient without stating why.

02.6 Around 75% of students answered this well. Incorrect answers included using non-standard conditions.

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QUESTION 03

03.1 Around 10% of students gained full marks on this question. Many students knew that accuracy and care were necessary but didn't explain what was required. Some knew the phrase 'weigh by difference' but then described washing the weighing bottle. Distilled (or deionised) water was required the first time it was used but this detail was not required later. Most students knew about transferring with washings, making up to the mark and mixing.

A few students followed the simple route of weighing out the solid and adding it to 250 cm^3 of water.

03.2 Around 20% of students gained full marks on this question.

In the first part, many students referred to 'avoid contamination'.

In the second part, simply stating 'remove bubbles' was insufficient. The focus had to be on the tap or jet of the burette.

Many students understood that some solution might drip from the funnel but found it hard to explain. Some just stated that it should be removed since it was no longer needed.

03.3 Some took 'minimum' to mean the start of the bottom curve of the end point. These students missed either one or both marks here, but could access 03.4 and 03.5 with their values.

03.4 This was generally done very well. A few students made transcription errors in their calculation.

03.5 Many students did very well on this question.

QUESTION 04

Most students answered this correctly.

QUESTION 05

Almost all students answered this question correctly.

QUESTION 06

Most students answered this question correctly.

QUESTION 07

This was answered well by many students.

QUESTION 08

Almost all students answered this question correctly.

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QUESTION 09

This was answered well by many students.

QUESTION 10

Although answered well, the three distractors were all popular.

QUESTION 11

This was done well, with option A the most popular distractor.

QUESTION 12

This was done moderately well but distractor B was popular.

QUESTION 13

Most students answered this well.

QUESTION 14

Most students answered this well, but options A, B and C were more or less equally popular distractors.

QUESTION 15

Most students answered this well.

QUESTION 16

This was answered well but a significant number of students chose option D.

QUESTION 17

This was done well but around one-third of students chose option D.

QUESTION 18

Almost all students answered this question correctly.

QUESTION 19

Around 60% of students answered this question correctly. Option D was a significant distractor.

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QUESTION 20

Most students answered this well, but options A and B were reasonably popular distractors.

QUESTION 21

Most students answered this well, but options B and C were popular distractors.

QUESTION 22

Most students answered this well, but options B and C were popular distractors.

QUESTION 24

Almost all students answered this correctly.

QUESTION 25

This was the least-well-answered question in Section B. Although the correct answer was the most popular, option D was a very significant distractor.

QUESTION 26

Most students answered this correctly.

QUESTION 27

Most students answered this well, but options A and D were popular distractors.

QUESTION 28

Around 70% of students answered this question correctly. Option A was a significant distractor.

QUESTION 29

Most students answered this question correctly.

QUESTION 30

This was answered well by over 90% of students.

QUESTION 31

Most students answered this correctly.

QUESTION 32

Most students answered this correctly.

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QUESTION 33

Most students answered this well, but option B was a popular distractor.

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